

# Answer Key & Teacher Notes

Session 1 – Writing Repeated Factors as Powers • Session 2 – Finding the Value of a Power

## HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

|                                  |                                                                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Warm-Up</b>                   | First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.                                                                                                                             |
| <b>Prerequisite Practice A/B</b> | Before the lesson, for students who need the skill the session uses. Key in that folder.                                                                                           |
| <b>Guided Notes</b>              | During the lesson. Students fill in the blanks as you teach; answers below.                                                                                                        |
| <b>Practice Form A</b>           | In class, after the lesson. Two pages. The self-check code tells students if they are on track.                                                                                    |
| <b>Practice Form B</b>           | Same skills, new numbers: reteach, retake, small group or absent students.                                                                                                         |
| <b>Exit Ticket</b>               | Last 5 minutes. Two per page; cut in half.                                                                                                                                         |
| <b>Homework Packet</b>           | Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support. |

## SESSION 1 • TEACHER NOTES

Writing Repeated Factors as Powers

### TALK PROMPTS

- Why do we say a power shows repeated multiplication and not repeated addition?
- Your partner says the exponent counts the multiplication signs. Is that true? What does it really count?

### WATCH FOR

- Switching the base and exponent ( $4 \times 4 \times 4 \times 4 \times 4$  written as  $5^4$ ).
- Using the number of  $\times$  signs as the exponent ( $8 \times 8 \times 8$  written as  $8^2$ ).
- Reading  $12^4$  as  $12 \times 4$ .

### PARTNER TASK

Write and Expand. One partner writes  $7 \times 7 \times 7 \times 7$  as a power and names the base; the other checks by expanding it back. Switch roles with  $4 \times 4 \times 4 \times 4 \times 4$ , then compare  $4^3$  and  $3^4$  in expanded form.

## SESSION 2 • TEACHER NOTES

Finding the Value of a Power

### TALK PROMPTS

- What pattern do you see in the powers of 2 as the exponent grows by one? Why does it happen?
- $4^4$  has the smaller base and  $9^3$  has the larger base. Which value is greater, and why?

### WATCH FOR

- Multiplying the base by the exponent ( $10^3 = 30$ ).
- Placing the decimal point wrong in decimal powers ( $0.6^2 = 3.6$  instead of  $0.36$ ).
- Thinking the larger base always gives the larger value ( $5^2$  vs.  $2^5$ ).

### PARTNER TASK

Storage Cube and Decimal Powers. One partner expands  $4^3$  and explains why it is not  $4 \times 3$ ; the other evaluates it (64). Switch roles to evaluate  $1.2^2$  (1.44) and  $0.3^3$  (0.027), checking decimal places.

## ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

| Sheet              | Standard | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
|--------------------|----------|---|---|---|---|---|---|---|---|---|----|----|
| <b>S1 • Form A</b> | 6.EE.A.1 | 1 | 2 | 1 | 1 | 2 | 2 | 3 | 2 | 3 | 1  | 1  |
| <b>S1 • Form B</b> | 6.EE.A.1 | 1 | 2 | 1 | 1 | 2 | 2 | 3 | 2 | 3 | 1  | 1  |
| <b>S2 • Form A</b> | 6.EE.A.1 | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 3 | 3 | 1  | 1  |
| <b>S2 • Form B</b> | 6.EE.A.1 | 1 | 2 | 1 | 1 | 2 | 2 | 2 | 3 | 3 | 1  | 1  |

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

## GUIDED NOTES

**power:** A short way to write the same **factor** multiplied again and again.

**base:** The number that is **repeated** as a factor.

**exponent:** The raised number that tells how many **times** the base is a factor.

- The **base** is the repeated factor.
- The **exponent** tells how many times the base is used as a factor.
- The exponent counts factors, not **multiplication** signs.

**Example:** The repeated factor is **2**. That is the base. Count the factors: there are **4**. That is the exponent. Write the base with a raised exponent:  $2^4$ . Answer:  $2^4$ , read “2 to the **fourth** power”

## YOU TRY

1. a.  $6^3$  b.  $(1/4)^2$
2.  $10 \times 10 \times 10 \times 10$ ; 4 factors

## EXIT TICKET

1. a.  $8^5$  b.  $0.3^2$
2. base 9, exponent 6;  $9 \times 9 \times 9 \times 9 \times 9 \times 9$
3. **B** ( $4^3$ )

## PRACTICE FORM A

1. a.  $7^4$  b.  $4^5$  c.  $5^3$  d.  $8^2$  e.  $(3/4)^2$  f.  $0.4^3$
2. a.  $2^6$  b.  $10^2$  c. base 2, exponent 6 d. The exponent counts how many times 2 is used as a factor.
3.  $5^4$ ; base 5, exponent 4
4.  $13^3$ ; base 13, exponent 3
5. Rafi is wrong. There are three factors of 8, so the exponent is 3:  $8 \times 8 \times 8 = 8^3$ .
6. a.  $(1/2)^3$  b.  $11 \times 11 \times 11$
7. a.  $2^5$  b.  $2 \times 2 \times 2 \times 2 \times 2 = 32$  layers c. Each fold adds another factor of 2, so the exponent grows by 1 and the layers double each time.
8. a.  $4^3 = 4 \times 4 \times 4$ ;  $3^4 = 3 \times 3 \times 3 \times 3$  b.  $3^4$  (it has 4 factors;  $4^3$  has 3)
9. The exponent 4 means use 12 as a factor 4 times, not multiply 12 by 4.  $12^4 = 12 \times 12 \times 12 \times 12$  (which equals 20,736).
10. **B** ( $3^5$ )
11. **A** (6)

**Self-check word** EXPONENT

## PRACTICE FORM B

1. a.  $6^3$  b.  $9^4$  c.  $11^2$  d.  $5^5$  e.  $(2/3)^3$  f.  $0.7^2$
2. a.  $3^5$  b.  $12^2$  c. base 3, exponent 5 d. The exponent counts how many times 3 is used as a factor.
3.  $4^6$ ; base 4, exponent 6
4.  $15^4$ ; base 15, exponent 4
5. Lena is wrong. There are four factors of 6, so the exponent is 4:  $6 \times 6 \times 6 \times 6 = 6^4$ .
6. a.  $(1/5)^2$  b.  $20 \times 20 \times 20$
7. a.  $3^4$  b.  $3 \times 3 \times 3 \times 3 = 81$  bacteria c. Each period adds another factor of 3, so the exponent grows by 1 and the count triples each time.
8. a.  $2^5 = 2 \times 2 \times 2 \times 2 \times 2$ ;  $5^2 = 5 \times 5$  b.  $2^5$  (it has 5 factors;  $5^2$  has 2)
9. The exponent 5 means use 3 as a factor 5 times, not multiply 3 by 5.  $3^5 = 3 \times 3 \times 3 \times 3 \times 3$  (which equals 243).
10. **B** (3)
11. **A** (times the base is used as a factor)

**Self-check word** MULTIPLY

## GUIDED NOTES

**evaluate:** Find the one **number** an expression equals.

**squared:** To the 2nd power; use the base as a factor **2** times.

**cubed:** To the 3rd power; use the base as a factor **3** times.

- To evaluate a power, write the **expanded** form and multiply one factor at a time.
- $4^3 = 4 \times 4 \times 4 = 64$ , not  $4 \times 3 = 12$ .
- Fraction and decimal bases work the **same** way:  $(1/2)^2 = 1/4$ .

**Example:** Expand:  $6^3 = 6 \times 6 \times 6$ . Multiply the first two:  $6 \times 6 = 36$ . Multiply by the last 6:  $36 \times 6 = 216$ . Answer:  $6^3 = 216$ , not  $6 \times 3 = 18$

## YOU TRY

- a. 81 b. 0.25
- $3 \times 3 \times 3 = 27 \text{ cm}^3$

## EXIT TICKET

- a. 25 b. 16
- $6 \times 6 \times 6 = 216 \text{ in}^3$
- C** ( $3^3$ )

## PRACTICE FORM A

- a. 144 b. 27 c. 16 d. 121 e. 1.44 f.  $1/9$
- $2^1 = 2 = 2$ ;  $2^2 = 2 \times 2 = 4$ ;  $2^3 = 2 \times 2 \times 2 = 8$ ;  $2^4 = 2 \times 2 \times 2 \times 2 = 16$ . Each value is 2 times the one before, because each new power has one more factor of 2.
- $4 \times 4 \times 4 = 64$  cubic units (not  $4 \times 3 = 12$ )
- $5 \times 5 \times 5 \times 5 = 625$
- The poster is wrong.  $7^2 = 7 \times 7 = 49$ , not  $7 \times 2 = 14$ .
- $2^5 = 32$ ,  $5^2 = 25$ ,  $3^3 = 27$ ; order:  $5^2$ ,  $3^3$ ,  $2^5$
- a.  $0.6 \times 0.6 = 0.36$  b.  $1.5 \times 1.5 = 2.25$
- a.  $3^4 = 81$  cells b.  $3^5 = 243$  cells c.  $243 - 81 = 162$  more cells
- Teo multiplied the base by the exponent.  $10^3$  means  $10 \times 10 \times 10 = 1,000$ .
- C** (32)
- A** (squared)

**Self-check word SQUARED**

## HOMEWORK PACKET • ANSWERS

*same answers for the core packet and Level 1*

## SHEET 1 • WRITING REPEATED FACTORS AS POWERS

- a.  $4^3$  b.  $9^2$  c.  $2^7$  d.  $10^4$  e.  $(1/3)^3$  f.  $0.5^2$  g.  $6^5$  h.  $(2/5)^2$
- Row 1: base 8, exponent 3,  $8 \times 8 \times 8$ . Row 2:  $7^4$ ,  $7 \times 7 \times 7 \times 7$ . Row 3:  $11^2$ , base 11, exponent 2.
- $3^5$
- He switched the base and exponent. The repeated factor 4 is the base, and it is used 5 times, so the power is  $4^5$ .

## PRACTICE FORM B

- a. 81 b. 8 c. 25 d. 169 e. 1.69 f.  $4/25$
- $3^1 = 3 = 3$ ;  $3^2 = 3 \times 3 = 9$ ;  $3^3 = 3 \times 3 \times 3 = 27$ ;  $3^4 = 3 \times 3 \times 3 \times 3 = 81$ . Each value is 3 times the one before, because each new power has one more factor of 3.
- $5 \times 5 \times 5 = 125$  cubic units (not  $5 \times 3 = 15$ )
- $6 \times 6 \times 6 = 216$
- The poster is wrong.  $8^2 = 8 \times 8 = 64$ , not  $8 \times 2 = 16$ .
- $3^4 = 81$ ,  $4^3 = 64$ ,  $2^7 = 128$ ; order:  $4^3$ ,  $3^4$ ,  $2^7$
- a.  $0.4 \times 0.4 = 0.16$  b.  $2.5 \times 2.5 = 6.25$
- a.  $4^3 = 64$  tiles b.  $4^4 = 256$  tiles c.  $256 - 64 = 192$  more tiles
- Malik multiplied the base by the exponent.  $9^3$  means  $9 \times 9 \times 9 = 729$ .
- D** (1,000)
- C** (36)

**Self-check word FACTORS**

## SHEET 2 • FINDING THE VALUE OF A POWER

- a. 49 b. 81 c. 125 d. 100,000 e. 64 f. 1
- a.  $1/8$  b. 0.09
- a.  $3^3$  b.  $10^4$
- $7 \times 7 \times 7 = 343 \text{ in}^3$
- A and C:  $2^6 = 64$  and  $4^3 = 64$  ( $6^2 = 36$ ,  $3^4 = 81$ )
- C** (64)
- A** ( $10^3$ )